BONE PATHOLOGY I

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HEREDITARY BONE DISORDERS

A. Achondroplasia

- Most common form of **inherited dwarfism**
- **(AD)-**Autosomal dominant (80% sporadic new mutations)
- Disorder of cartilage proliferation
- Mutation is : Overactive FGFR3 **inhibits** cartilage proliferation at epiphyseal growth plate, resulting in decreased endochondral bone formation

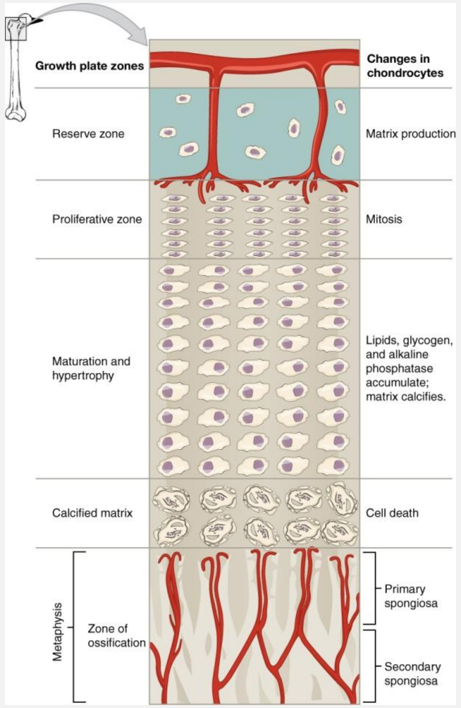
Pathogenesis

- During <u>embryonic</u> establishment of the growth plate before formation of the secondary ossification center, ***FGFR3 signaling promotes* chondrocyte proliferation**

- However, during <u>postnatal</u> skeletal growth, *FGFR3 signaling* ***inhibits*** chondrocyte proliferation and differentiation

- *In Achondroplasia FGFR3 is* ***constitutively active***

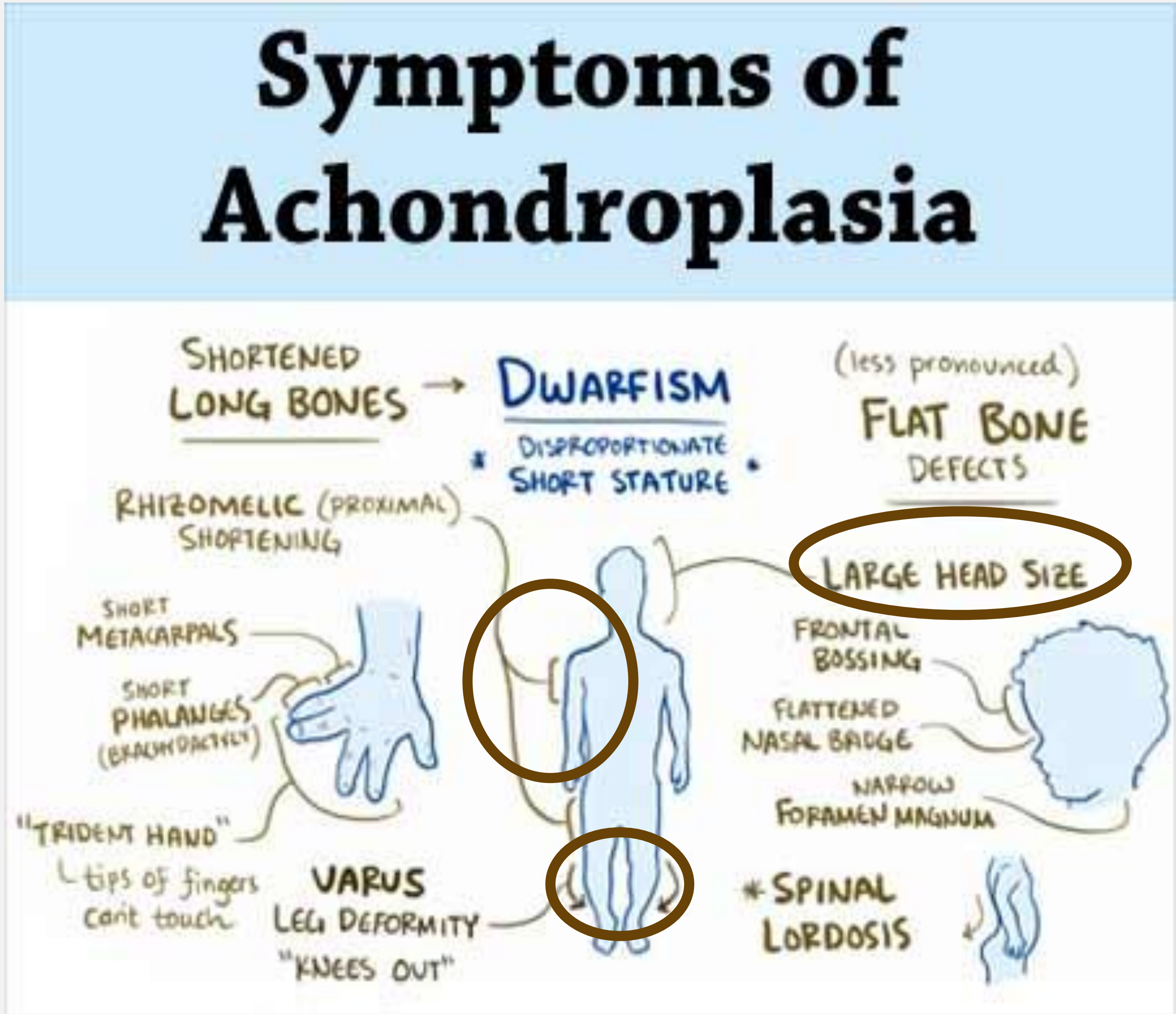
Activated FGFR3
Inhibits proliferation



Diagnosis

The diagnosis of Ach is usually **<u>made at birth</u>**

- **short long bones**, disproportional shortening of the <u>proximal skeletal segments</u> **(rhizomelia)**,
- **tibial bowing,** exaggerated lumbar lordosis, ,
- **macrocephaly,**
- **hearing loss**



B. Osteogenesis imperfecta (brittle bone disease)

- Autosomal Dominant
- Abnormal synthesis **of type I collagen and bone mineralization**

- Increased susceptibility to bone fractures
- Decreased bone density.

Skeletal:
short stature,
scoliosis,
skull deformities

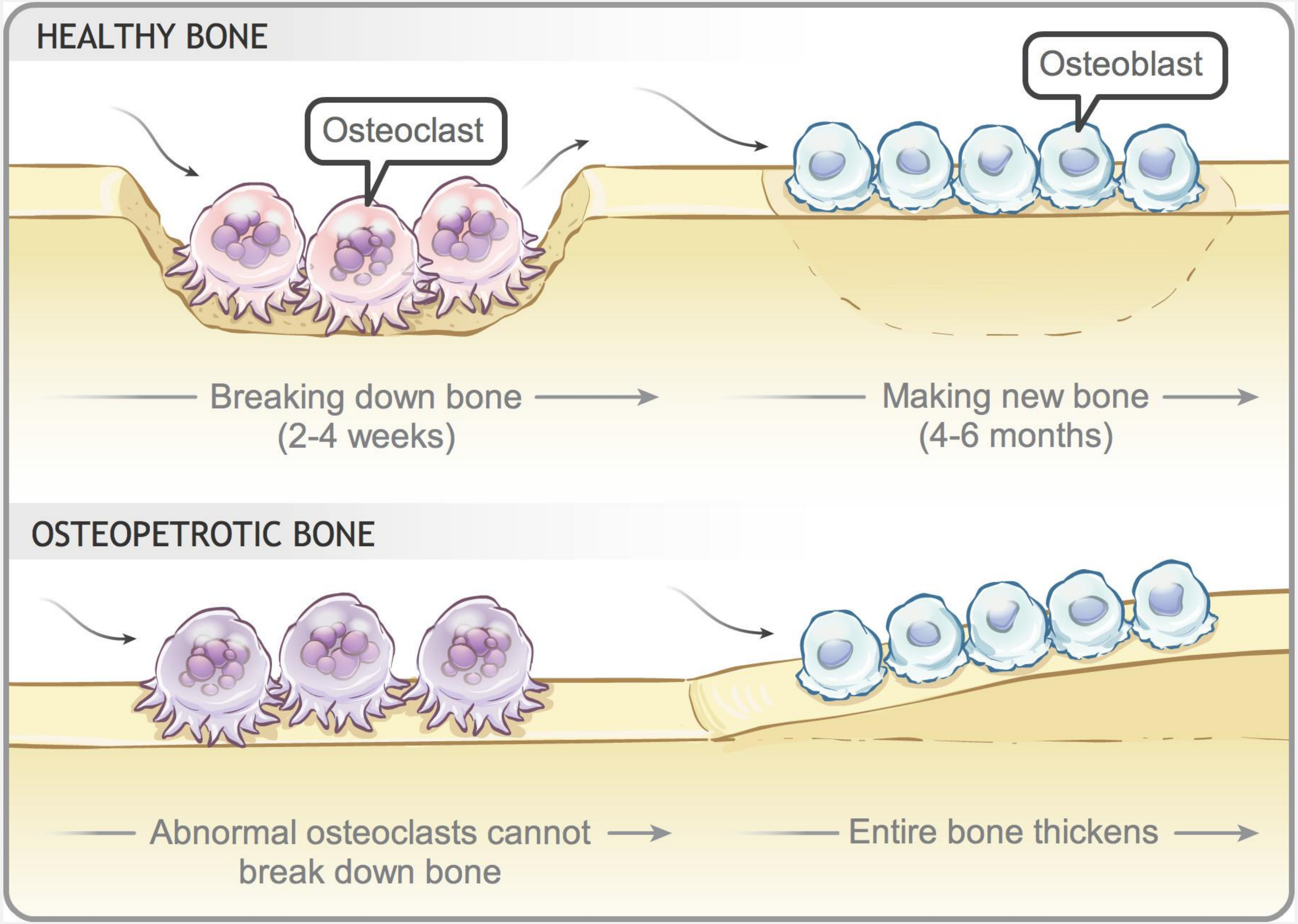
Extraskeletal:
hearing loss
sclera may be blue
joint hyperextensibility

C. Osteopetrosis

Osteopetrosis is a heterogeneous group of **heritable conditions** in which there is a **defect in bone resorption by osteoclasts.**

Bone is a **dynamic tissue** in which osteoblasts synthesize bone matrix while *osteoclasts resorb bone*.

Bone **density** is dependent on the relative function of these two types of cells. Osteoclasts and Osteoblasts


HEALTHY BONE
Osteoclast
Osteoblast
Breaking down bone (2-4 weeks)
Making new bone (4-6 months)
OSTEOPETROTIC BONE
Abnormal osteoclasts cannot break down bone
Entire bone thickens

Pathology:

- Impaired **osteoclast's** function,

- Decreased **resorption** leads to **thick sclerotic bones**

- Interference of **acidification** by osteoclasts , required for bone resorption.

- Defective gene which encodes for **carbonic anhydrase II and chloride channel gene** (*acidic environment is necessary to remove bone)*

Pathogenesis and basics in depth:

Osteoclast precursors are of **monocyte lineage** which differentiate into osteoclasts. Osteoclasts become adherent to bone.

A tight seal of attachment between podocyte of osteoclast and bone surface .

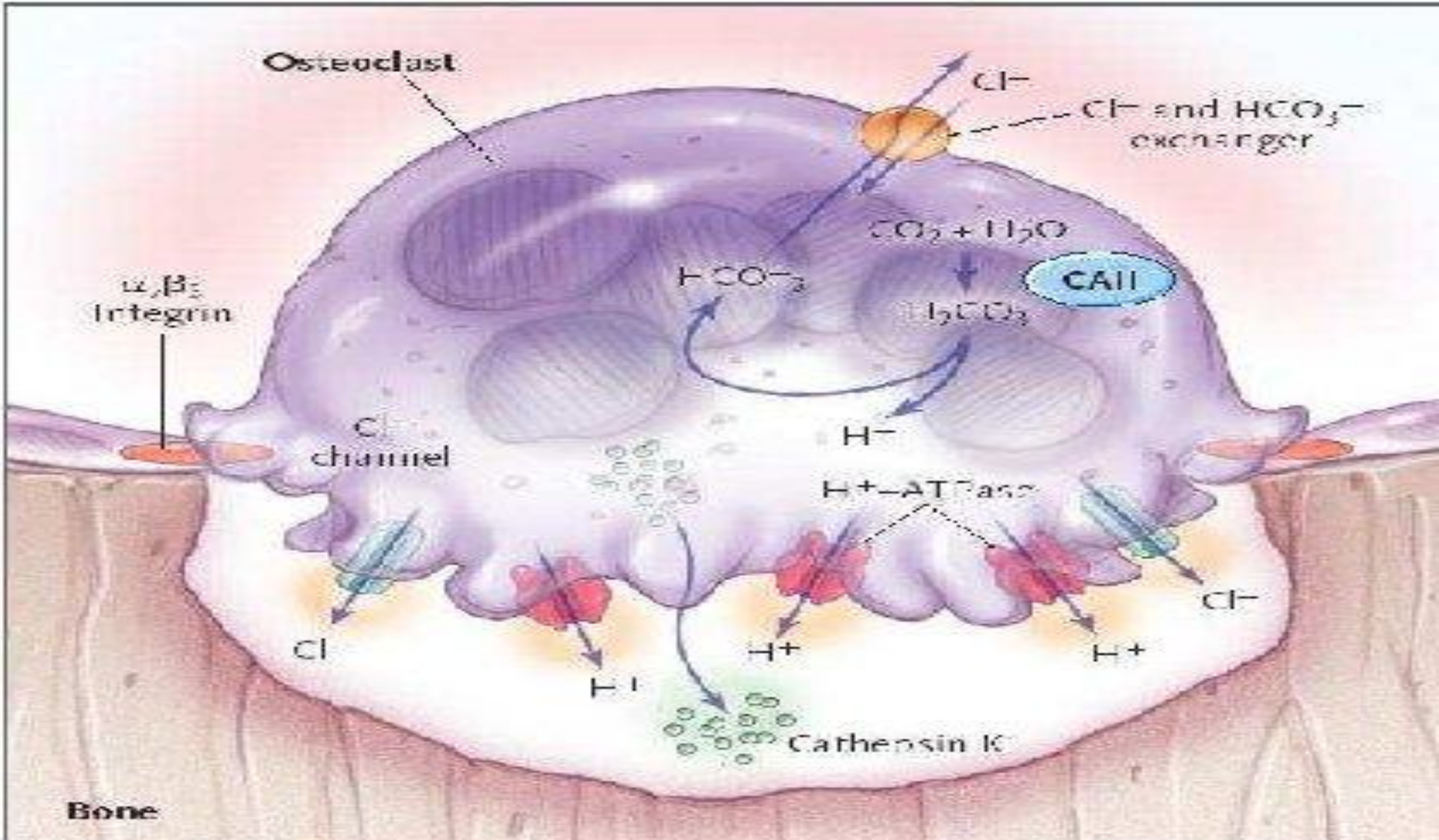
Polarization of the osteoclast forms two functionally distinct domains, the resorptive surface and the basolateral membrane .The **"ruffled border,"** is the resorptive surface of the cell.

Resorption occurs through **acidification of the bony surface**, which initiates dissolution of the mineral matrix and secretion of enzymes that digest the organic component of bone.

In the **cytoplasm of the osteoclast**, **carbonic anhydrase II** forms carbonic acid (H_2CO_3) from carbon dioxide (CO_2) and water; the H_2CO_3 dissociates to bicarbonate (HCO2 ¡) and a proton (H^+).

The protons are transported through the ruffled border into the resorption lacuna by a vacuolar proton pump (H^+–ATPase), **generating a pH of 4 to 5** in the extracellular space adjacent to bone.

Acidification of this extracellular environment initiates the degradation of the mineral component of bone, which is composed primarily of hydroxyapatite



Osteoclast
Cl−
Cl− and HCO3− exchanger
CO2 + H2O
CAII
HCO−3
H2CO3
αvβ3 Integrin
H+
Cl− channel
H+–ATPase
Cl−
Cl
H+
H+
H+
Cathepsin K
Bone

HEREDITARY DISORDERS OF BONE

ACHONDROPLASIA	**OSTEOGENEISIS IMPERFECTA**	**OSTEOPETROSIS**
decreased endochondral bone formation	Abnormal synthesis of type I collagen and bone mineralization	Impaired osteoclasts